

Mohammad Montasim Al Mamun Shuvo

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SKILLS

Frontend: React.js, Next.js, TypeScript, Tailwind CSS, Redux Toolkit, WebRTC

Backend & Data: Node.js, Express.js, WebSocket, REST APIs, PostgreSQL, MongoDB, Prisma

Cloud & DevOps: Microsoft Azure, Docker, GitHub Actions, CI/CD

Testing & Tools: Jest, React Testing Library, Lighthouse, Figma, Git

Specialized: Real-time video processing, Canvas-based annotation, SSO, and System Design

EXPERIENCE

MyMedicalHub International Ltd.

Mohakhali, Dhaka, Bangladesh

Health-Tech SaaS providing AI-driven biometric assessments and telemedicine.

- **Senior Software Engineer** Oct 2025 – Present
 - Architected an FSM-based biometric engine, replacing unstable React hooks with deterministic state transitions to ensure 99.9% reliability during AI analysis.
 - Optimized AI pipelines to support 60 FPS pose estimation and real-time rep counting via adaptive frame rate logic.
 - Led a full architectural rewrite of the medical assessment chatbot with Redux Toolkit that increased form completion rates.
 - Refactored 54+ modules into a Presentational/Container pattern, decoupling business logic from the UI.
 - Eliminated memory leaks and state bottlenecks by designing imperative recording controls and MediaRecorder instance.
- **Software Engineer** Oct 2023 – Sep 2025
 - Boosted application performance by 40% by stripping 4 legacy UI libraries and optimizing React rendering cycles.
 - Achieved a 70% infrastructure cost reduction by migrating Azure VMs to Azure App Service and automating CI/CD pipelines.
 - Deployed a HIPAA-compliant SSO system with multi-role access, OAuth integration, and strict CSP headers.
 - Engineered an automated clinical reporting engine with HTML/PDF rendering for branded B2B medical data visualization.
- **Junior Software Engineer** Sep 2022 – Sep 2023
 - Reduced diagnosis time by 25% and achieved 95% accuracy by integrating real-time AI analysis and custom algorithms.
 - Integrated a WebRTC video pipeline via Opentok, stabilizing remote consultations and reducing call latency.
 - Secured core data transmission by developing a JWT-based REST API client layer with built-in XSS and CSRF protection.
 - Accelerated initial app load times through lazy loading and route-based code splitting across the application.

PROJECTS

MyMedicalHub International Ltd.

Dhaka, Bangladesh

Core technical contributions to the platform's AI-driven healthcare ecosystem.

- **Patient Portal**
 - Built a Biometric Movement Engine from scratch using MediaPipe, achieving 60 FPS adaptive pose estimation.
 - Led a Redux Toolkit rewrite of the Chatbot Assessment Module, implementing dynamic question routing.
 - Designed a multi-tenant feature flag system and API-driven dynamic theming to manage 250+ client configurations.
 - Optimized performance by replacing 5 legacy UI libraries with custom high-performance components.
 - Technology: **React.js, Redux Toolkit, MediaPipe, Microsoft Azure, Google Tag Manager**
- **Telemedicine Microservice**
 - Architected a Next.js application for remote physical therapy, utilizing state machines for robust WebRTC peer management and recording lifecycles.
 - Built P2P video infrastructure using STUN/TURN servers and the MediaRecorder API for secure session archival.
 - Engineered a bidirectional event system via Socket.IO to orchestrate real-time clinical feedback.
 - Managed full CI/CD and production stability across multi-environment deployments ensuring high availability.
 - Technology: **Next.js, TypeScript, Socket.IO, AWS S3, Tailwind CSS, PM2**
- **Re-Annotation Microservice**
 - Developed a Canvas-based post-processing engine for correcting biometric ROM data, featuring high-precision drawing primitives and affine transformations using Konva.js with pixel-accurate frame comparisons.
 - Secured the SPA with enterprise-grade protocols, including CSP headers, XSS sanitization, and CSRF mitigation layers.
 - Optimized image processing via dynamic resizing and aspect ratio normalization for seamless experience.
 - Technology: **Konva.js, Node.js, Express.js, JWT, HTML5 Canvas, REST APIs**

EDUCATION

- **Bangladesh Army University of Science and Technology** November 2016 — March 2021
B.Sc. in Computer Science & Engineering